SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security COURSE CODE: CSA5116

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:50**

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I Ganesh. B / 192511066 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: B. Ganesh

CSA5116 - Cryptography and Network security

CO1 AT1 - Application Based Questions.

21) Event communication Security :-

Scenario :-

A student organization wants to securely share the message "MEET AT NOON" using a fixed shift encryption method.

a) Encryption using caesar cipher with shift 4 :-

The ceaser shifts every letter 4 positions forwarded in the alphabet

M	E	E	T	A	T	N	O	O	N
+4 shift	Q	I	I	X	E	X	R	S	S	R

Therefore :-

plaintext :-

MEET AT NOON

ciphertext :-

QII XEX RSSR

spaces are normally retained for readability.

b) Decryption :-

An authorized member who knows that the shift value is 4 can decrypt the message by shifting every ciphertext letter 4 positions backward.

Example :

Q → M
I → E
I → E
X → T
E → A
X → T
R → N
S → O
S → O
R → N

Therefore :

ciphertext : QI IX EX RSSR

Decryption : MEET AT NOON

Application :

This Method can be used for simple Educational or low security communication.

conclusion :

The Caesar cipher successfully converts the original Message into an unreadable form using a fixed shift, and the authorized receiver can recover it using the reverse shift.

1) A student organization is preparing to share confidential event details with its core Members using a simple classical encryption Method. During Testing, a Message "MEET AT NOON" is into an encoded version

a) Reconstruct how the encrypted Message might appear if a shift of 4 positions is applied.

b) Explain how an authorized Member, aware of the enlodiing rule, would retrieve the original Message.

4) Suspicious Digital Asset Investigation :-

During a Digital forensic audit, Investigat-ors suspect that a company logo file may contain hidden confidential Information embedded within it.

a) Analyze how Investigators could detect the presence of hidden data within the image.

b) Recommend two techniques that could strengthen the security of such communication against detection or extraction.

Scenario :-

During a forensic Investigation, a company logo is suspected of containing hidden confidential Information. The Image looks normal during casual Inspection.

a) Detecting hidden Data :-

Investigators can use steganography detection techniques to determine whether secret Information is hidden inside the Image.

1) Visual Inspection :-

The Investigater first checks the Image for :-

- unusual pixels
- strange color variations
- Disorted areas
- unexpected patterns

2) File size analysis :-

The image file size is compared with a normal image of similar resolution and quality.

If the logo has an unusually large file size, it may indicate hidden information.

3) Metadata Analysis :-

Investigators examine Metadata such as

- File creation date
- Modification date
- software used
- Comments
- Author Information.

4) Statistical Analysis :-

Analyze pixel patterns for abnormalities

5) steganography tools - use forensic tools to detect embedded data.

6) Two security Techniques :-

1. Encrypt the message first - Even if the hidden data is extracted, it cannot be understood without the key.

2. Use adaptative steganography - Hide data in areas where changes are difficult to notice.

conclusion :- combining encryption + steganography provides better security for confidential communication.

